Counterexamples to both Bernoulli–Toda denominator claims

Principia Math In Progress

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Task 1: MathDB 371655. For an even positive integer N, let

$$P(N) = \{p \text{ prime} : p - 1 \mid N\}, \qquad D_N = \prod_{p \in P(N)} p, \qquad F(d) = \min\{N > 0 : N \text{ even},\ D_N = d\}.$$

The formula for the denominator D_N of the Bernoulli number B_N is the classical von Staudt–Clausen theorem, used in the source [1, Section 5.1]. Its Toda primes are

$$T(n) = \left\{ p \text{ odd prime} : p - 1 \mid 4n,\ \gcd\left(p, \frac{4n}{p-1}\right) = 1 \right\}, \qquad t(n) = |T(n)|.$$

Conjecture 5.6 of [1] asserts that, for a Bernoulli denominator d, (i) if $F(d) = 4a$ and $D_{4am} = D_{4a} = d$, then $T(am) = T(a)$; and (ii) if $D_{4am} \neq D_{4a} = d$, then $t(am) \geq t(a)$. Our second example also satisfies $F(d) = 4a$, whether or not that condition is carried over to clause (ii).

Theorem 1. *Both clauses of Conjecture 5.6 are false.*

(i) $a = 23,\ m = 47,\ d = 1410 :$ $\quad F(d) = 92, \quad D_{4324} = D_{92} = 1410,$
$T(23) = \{3, 5, 47\}, \qquad T(1081) = \{3, 5\};$

(ii) $a = 58,\ m = 13747 = 59 \cdot 233,\ d = 412410 :$ $\quad F(d) = 232,$
$D_{3189304} = 192595470 = 467 D_{232} > D_{232},$
$T(58) = \{3, 5, 59, 233\}, \qquad T(797326) = \{3, 5, 467\}.$

An elementary reduction. For an odd prime p with $p - 1 \mid 4n$, the condition $p \mid 4n/(p-1)$ is equivalent to $p \mid n$, since $\gcd(p, p-1) = \gcd(p, 4) = 1$. Thus, writing $\Omega(n)$ for the prime divisors of n,

$$T(n) = P(4n) \setminus \big(\{2\} \cup \Omega(n)\big). \tag{1}$$

This equivalent definition already appears in the source and in the 2018 MathOverflow discussion [3]. We use it only to simplify the finite arithmetic.

Proof of (i). Since $92 = 4 \cdot 23$, its even divisors are $2, 4, 46, 92$. Adding one gives $3, 5, 47, 93$, of which exactly the first three are prime. An odd divisor plus one is an even number, prime only for divisor one. Consequently $P(92) = \{2, 3, 5, 47\}$ and $D_{92} = 1410$. If $D_N = 1410$, both 5 and 47 divide D_N, so $4 \mid N$ and $46 \mid N$. Hence $92 = \operatorname{lcm}(4, 46) \mid N$, proving $F(1410) = 92$.

Now $4324 = 4 \cdot 23 \cdot 47$. Relative to 92, the additional even divisors are $94, 188, 2162, 4324$. Each gives a composite successor:

$$95 = 5 \cdot 19, \quad 189 = 3 \cdot 63, \quad 2163 = 3 \cdot 721, \quad 4325 = 5 \cdot 865.$$

The old composite successor is $93 = 3 \cdot 31$. These exhaust all even divisors, so $P(4324) = P(92)$ and $D_{4324} = D_{92}$. Equation (1), with $1081 = 23 \cdot 47$, gives $T(23) = \{3, 5, 47\}$ but $T(1081) = \{3, 5\}$, disproving (i).

Proof of (ii). We have $232 = 8 \cdot 29$. Its even divisors are $2, 4, 8, 58, 116, 232$, whose successors are $3, 5, 9, 59, 117, 233$. Hence

$$P(232) = \{2, 3, 5, 59, 233\}, \qquad D_{232} = 2 \cdot 3 \cdot 5 \cdot 59 \cdot 233 = 412410.$$

Any N with this denominator has $233 - 1 = 232 \mid N$. Thus $F(412410) = 232$.

Set $n = 58 \cdot 59 \cdot 233 = 797326$, so $4n = 2^3 \cdot 29 \cdot 59 \cdot 233$. Every even divisor of $4n$ is $2^j u$ with $j \in \{1, 2, 3\}$ and

$$u \in \{1, 29, 59, 233, 1711, 6757, 13747, 398663\}.$$

The following table lists *every* such divisor's successor. A bold entry is prime; every other entry has an explicit proper factor.

u	$2u+1$	$4u+1$	$8u+1$
1	**3**	**5**	$9 = 3 \cdot 3$
29	**59**	$117 = 3 \cdot 39$	**233**
59	$119 = 7 \cdot 17$	$237 = 3 \cdot 79$	$473 = 11 \cdot 43$
233	**467**	$933 = 3 \cdot 311$	$1865 = 5 \cdot 373$
1711	$3423 = 3 \cdot 1141$	$6845 = 5 \cdot 1369$	$13689 = 3 \cdot 4563$
6757	$13515 = 3 \cdot 4505$	$27029 = 151 \cdot 179$	$54057 = 3 \cdot 18019$
13747	$27495 = 3 \cdot 9165$	$54989 = 11 \cdot 4999$	$109977 = 3 \cdot 36659$
398663	$797327 = 41 \cdot 19447$	$1594653 = 3 \cdot 531551$	$3189305 = 5 \cdot 637861$

For the entries declared prime, trial division by primes at most $\sqrt{467} < 22$ suffices; none of $2, 3, 5, 7, 11, 13, 17, 19$ divides 467, and the same test up to the appropriate square root proves primality of 59 and 233. The odd divisors contribute only prime 2. We have therefore proved

$$P(3189304) = \{2, 3, 5, 59, 233, 467\}, \qquad D_{3189304} = 467 D_{232} = 192595470.$$

Since $\Omega(58) = \{2, 29\}$ and $\Omega(797326) = \{2, 29, 59, 233\}$, equation (1) now gives the two claimed Toda sets. In particular $t(797326) = 3 < 4 = t(58)$, disproving (ii), even with the minimum-index condition imposed. □

Scope and verification. These examples disprove both clauses of the stated denominator conjecture. They do not refute the separate conjecture $t(n) \geq 2$ for every positive integer, or the source's restricted conjecture about multiples of 3. All displayed numbers and hypotheses are exact; no transcription or omitted-hypothesis issue is used. The elementary characterization of Toda primes, von Staudt–Clausen, and minimum-index divisibility are classical or source material, not claimed as new.

The verifier checks all divisor successors and proper factors, independently sieves every prime up to the largest possible candidate and applies the original gcd definition, checks both minimum indices, and computes three smaller-index Bernoulli numbers exactly with FLINT. The large-index denominator is certified by the exhaustive prime and divisor calculations, without computing its enormous numerator. All 15 pages of the current target source were read; the author's 15-page copy has identical extracted text except for the arXiv stamp. No target journal final or earlier full target claim was identified in the recorded bounded audit. Review is by the authoring agent only.

Task 2: Classify the Bernoulli denominators d with $4 \mid F(d)$ for which the Toda set is constant on all indices N satisfying $D_N = d$.

Task 3: Determine whether clause (i) has infinitely many counterexamples with both a and m prime.

Task 4: Determine whether $t(am) - t(a)$ is unbounded below when $F(D_{4a}) = 4a$ and $D_{4am} > D_{4a}$.

Task 5: Find sufficient arithmetic conditions on a, m that recover $t(am) \geq t(a)$ despite the loss of primes dividing m.

References

[1] S. McKean, Toda primes, arXiv:2511.19744v1 (2025), Definition 1.1; Conjecture 5.6, p. 8.

[2] C. Pomerance and S. S. Wagstaff Jr., The denominators of the Bernoulli numbers, Acta Arith. 209 (2023), 1–15, doi:10.4064/aa210601-16-11. Proposition 1 gives the general identity $F(d) = \mathrm{lcm}_{p|d}(p-1)$.

[3] A. Milovanov, Number theory question from Homotopy groups of spheres, MathOverflow (15 April 2018), question 297915, including G. Martin's answer.